

APT to Disagree: A Comparative Analysis of Attribution in Commercial TI

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Abstract—Attributed cyber threat intelligence (TI) plays an important role in the effective mitigation of cyber attacks. Yet, despite the central role of attribution in policy, practice, and vendor reporting, little is known about the coverage and reliability of attribution by threat intelligence vendors. No study has systematically investigated attribution across a large set of leading TI vendors. We close this gap and provide a longitudinal comparative analysis across 13.5 million IOCs collected over the last 14 years from seven vendors. To compare IOC attribution across vendors, we normalize heterogeneous feeds and reconcile actor names using an evaluated and augmented version of MISP Threat Actor Galaxy (MISP TAG). Next, we address two questions: (i) what is the scope of actor-tracking by vendors, and (ii) how consistent is attribution among vendors? We find that the majority of actors tracked by one vendor are not tracked by the other. Furthermore, IOCs observed by multiple TI vendors are rare (1%), illustrating that commercial TI feeds, like open-source feeds, primarily provide singleton IOCs. We also find limited overlap in IOCs for jointly tracked actors by two vendors. We measure attribution agreement among vendors with Krippendorff’s α . We find mostly moderate agreement among vendors for actor attribution. By contrast, country attribution has high agreement. Our results have implications for actor-centric defenses, compliance, and geopolitical uses of attribution.

Index Terms—Threat Intelligence, Attribution, IOCs, APT

1. Introduction

Attributed Indicators of Compromise (IOCs) are pieces of forensic data that indicate a system or network may have been breached by a particular threat actor. They play an important role in high-end threat intelligence (TI), enabling defenders to detect, respond to, and anticipate threats more effectively, while also informing strategic and legal decisions. Unlike non-attributed counterparts, attributed IOCs provide organizations with insights into who is behind an attack and why it is taking place [1]–[4].

Attribution of threats to actors is critical to various use cases. First, best practices advocate that organizations adopt TI to know the adversaries targeting them, so that they can focus their resources on tailored defences [5]–[9], thereby

directing resource expenditures and improving defense effectiveness. This threat-led approach has also been incorporated into government and corporate policy guidance [2], [10], [11]. Second, attributed IOCs are used to assess legal exposure and sanction checks [12]. Some specific threat actors might be sanctioned by policy makers, such as the U.S. Office of Foreign Assets Control [13], and enterprises have to adapt their engagement (including ransom payments) and reporting obligations accordingly. Third, attribution has geopolitical implications [14]. Tying attacks to certain states or organizations has real-world consequences such as embargos on countries, export controls, sanctions, or exerting diplomatic pressure [15], [16].

Established models – Pyramid of Pain, the Q-model, and MITRE ATT&CK – differentiate short-lived, easily changed low-level IOCs (e.g., IPs, URLs) from more persistent high-level behavioral patterns, i.e., Techniques, Tools and Procedures (TTPs), typically treating the latter as more reliable for attribution [17]–[19]. However, recent studies show TTPs alone rarely discriminate actors, and practitioners often ground investigations in low-level indicators [12], [18], [20]–[22]. Consequently, low-level IOCs remain indispensable to contemporary attribution and the decisions derived from it.

Despite the central role of attribution in policy, practice and vendor reporting, little is known about the coverage and reliability of attribution by threat intelligence vendors. Disagreements in attribution risk undermining the very ‘threat-led’ defenses they are intended to support. While there is a wealth of prior work on threat intelligence, focusing on various aspects of quality like accuracy, timeliness, and coverage [23]–[25], attribution is rarely mentioned in this research and, to the best of our knowledge, has never been evaluated. This is likely related to the fact that the bulk of this work is based on open source TI, which lacks the attribution effort of high-end commercial vendors. Only one study [26] has analyzed the products of two leading threat intelligence vendors. It briefly touched on attribution and found little overlap in the IOCs of the two vendors when tracking the same actor. However, to date, no study has systematically investigated attribution across a large set of leading TI vendors.

In this paper, we aim to close this gap. We conduct a longitudinal comparative analysis of 13.5 million unique,

attributed IOCs collected over the last 14 years (2011-2025) from seven commercial TI feeds. Determining the quality of the attribution of IOCs is severely impeded by the lack of ground truth data on attackers that hampers all research on TI [24]. In addition to absent ground truth, the attribution process itself is confidential, using data and knowledge that vendors do not make public. That makes independent evaluations very difficult. Yet, what we can do is analyze and compare the attribution claims that different vendors make about their IOCs. Simply put, if these claims conflict, then at least one of those attributions is incorrect. We seek to answer two research questions – RQ1: *what is the scope of actor-tracking by vendors?* and RQ2: *how much agreement is there in attribution across vendors?*

We find that the majority of actors are tracked by only one vendor. When vendors do track the same actor, they report sets of IOCs with very low overlap. When multiple vendors report the same IOC, the level of agreement in their attribution is poor – as indicated by Krippendorff’s α , a metric for inter-rater reliability. Agreement levels improve to satisfactory when we remove temporary attribution labels. The agreement about which country is behind the attack is consistently high among vendors. In sum, we make the following contributions:

- We present the first systematic investigation of attribution in high-end commercial threat intelligence feeds, providing a longitudinal comparative analysis across 13.5 million IOCs of seven vendors.
- Recent work found that the lack of a shared industry mapping was hampering analysts [27]. We validate and augment the leading industry actor mapping, MISP Threat Actor Galaxy (MISP TAG or TAG), which contains 855 actors. While not exhaustive, our validation shows TAG has high accuracy when checked against a deconflicted ground-truth mapping from CrowdStrike and Microsoft and against vendor TI reports for a random sample of 50 actor names.
- We find that vendor coverage of threat actors is limited. Each vendor covers only a fraction of the actors (2-17%) in TAG. The union of all seven feeds, which would cost USD 1-2 million per year, is able to track just 34% of the actors from TAG.
- We find that only 1% of all attributed IOCs are observed by more than one vendor. Even when vendors claim to track the same threat actors, the IOCs they report for those actors show minimal overlap. When vendors do observe the same IOC, their attribution agreement is moderate: Krippendorff’s α ranges from 0.76-0.81 for two vendors, even when the observations are just a week apart in time. Agreement on country attribution is consistently high, with α ranging from 0.92-0.95 for two vendors. There is poor agreement on IPv4 indicators (α ranges from 0.65-0.71), but even on MD5 hashes, disagreement remains.
- We identify implications for security practices around the use cases of attributed TI.

2. Related Work

Our work resides at the nexus of studies on threat intelligence, attribution, and Advanced Persistent Threats (APTs).

Threat Intelligence. There is a rich line of work on, primarily, open-source TI. Early efforts focused on abuse feeds and block lists and found limitations in accuracy, timeliness, and coverage [23]–[25]. More recently, studies have described TI IOCs from open and commercial sources and measured them along various quality dimensions, including, among others, coverage, timeliness, relevance, overlap, delay, volume, and accuracy [26], [28]–[32]. Studies that measure the quality of TI all suffer from the lack of ground truth [24]. The authors of [33] conducted a qualitative study to identify the most effective ways to present TI quality to analysts. Bouwman et al. [26] conducted a mixed-method study to determine how customers value commercial TI. They found that TI wasn’t exclusively used for detection purposes but also to understand the threat landscape. To our knowledge, it is the only study analysing IOCs of reports from two leading commercial vendors. TI promises to enable the detection of attackers within defenders’ networks. It assumes that once an actor is observed in one network, it may be observed elsewhere [34]. Several studies examined the effectiveness of IOCs for timely intrusion detection and tried to determine the optimal lifespan of IOCs [35]–[38]. The authors of [38] found that IOC lifetimes were not static and developed a technique to dynamically estimate lifetimes.

Frameworks and models. Several models include the use of TI for cyber defense. The ‘pyramid-of-pain’ model illustrates that some indicators are harder to replace for attackers than others. Low-level IOCs can be quickly replaced, and TTPs are therefore believed to be the hardest to change. The Q-model of Thomas Rid relates to attribution and identifies three different levels: tactical, operational, and strategic. On the tactical, and most technical, level, Rid observes that low-level IOCs play a key role at the start of attribution investigations [18]. In [15], Steffens describes the fine-grained MICTIC framework for technical attribution. It is similar to the Diamond model, a framework to analyze cyber intrusions [39].

Attribution. Some prior work has focused on tools for (APT) attribution [40]–[42]. The authors of [43] proposed CSKG4APT, a knowledge graph model for attribution using Open-Source Cyber Threat Intelligence (OSCTI). In [44], [45], the authors applied machine learning to classify APT groups using features from sandbox reports and string and code features, respectively. Scrutinizer identified unknown APT samples through code reuse analysis [46]. In another study, the authors proposed ADAPT, which utilized static features extracted from heterogeneous file types for attribution by clustering executables and documents in threat groups and campaigns [47]. In [45], the authors developed explainable APT attribution for malware using NLP techniques.

There is another strand of research about attribution as a practice done in industry and in the context of international

relations [18], [48]–[50]. More recently, [27] shed light on the perceptions of security practitioners in analyzing APT-level threats, including attribution. Other attribution-related work focuses on empirical evaluations of various claims within industry. Recent studies [12], [20] showed that the use of TTPs to distinguish between threat actors has its limitations. In [51], the authors found that a big challenge in attribution is the unreliable naming and labeling of threat actors and their associated TTP. Leite et al. demonstrated that name server patterns may be useful for clustering attacks [22]. All these studies find that low-level IOCs remain essential for attribution, though some point to a mix of low and high-level IOCs [52], [53]. Indeed, respondents of a SANS survey indicated that low-level indicators of compromise were valued higher than information about TTPs [21]. **Attacker Economics and APTs.** Various studies looked into the operations of attributed attackers. In [54], [55], the authors found that most attack campaigns described in TI use known vulnerabilities. This supports the idea of [56] that, in general, attackers operate cost-efficiently. By contrast, the authors of [57] found that APT-attributed malware used more sophisticated evasion techniques.

In this work, we close research gaps in existing work by conducting an empirical and longitudinal study of seven feeds of commercial TI. Previously, only [26] studied two commercial TI feeds. However, no study has systematically investigated attribution across a large set of leading TI vendors.

3. Background

As threat actors carry out attacks, they leave traces within victim systems and networks. Security firms collect victim logs and analyze malicious artifacts to identify clusters and patterns. The attribution of attacks to specific actors is a dynamic, ongoing process, as vendors continually acquire new data and insights. These may originate from additional victim systems, other customer networks, collaborations with vendors, or publicly available sources such as forums and the dark web.

In [15], Steffens explains the role of ‘intrusion sets’ in the analysis and attribution of cyber attacks. Malicious activity can be analyzed by observing both low-level indicators and TTPs being used together repeatedly. Such co-occurrences are termed intrusion sets. One actor can be associated with more than one intrusion set – i.e., it can use different approaches for an attack. For example, a phishing e-mail attack versus a watering hole attack. Each attack has its own artefacts, malware, control servers, etc.

Security firms collect a catalog of intrusion sets against which they can compare new attacks. Historical data is of high value for correlating incidents [27]. If TTPs and/or indicators match or overlap, the probability that the same group is behind the attack increases. An intrusion set is typically assumed to refer to a single actor, even if the specific identity of the threat actor may still be unknown. This process explains why threat actor attribution is so dynamic and has a high degree of uncertainty. There is no

ground-truth about attackers, only telemetry data and probabilities [15]. For this reason, some vendors provide a type of ‘work in progress’ name or label, for intrusion sets that are not yet attributed to a specific named actor. Customers can use the “early attributed” IOCs for block lists, detection signatures, and patching priorities [58]. Vendors each use their own convention for these ‘work in progress’ (WIP) names. For example, Mandiant uses UNC[XXXX] (e.g., UNC1878), Recorded Future and Microsoft use TAG-[XX] and DEV-[XXXX], and Trend Micro uses VOID[X] [58], [59].

Vendors all use their own naming schemes for actors. For example, with the SolarWinds supply-chain intrusion in 2020, Mandiant tracked the actor as UNC2452, Microsoft as NOBELIUM, while CrowdStrike labelled the associated actor APT29 [60]. Security firms have their own customer base and therefore observe attacks that might or might not have been observed by other firms as well. This also means that some of their telemetry data is unique. Analysts at different security companies may focus on different aspects of a set of attacks, resulting in differently delineated intrusion sets. For this reason, the vendor-specific naming scheme, at least initially when uncertainty is high, makes sense.

Yet, vendors do not report on actors in a vacuum; they collaborate, and their analysts read and refer to other vendors’ TI reports [27]; thus, gaining insights from each other’s reporting. Consequently, in their own reports, they often provide the reference names (aliases) used by other vendors for that actor. This information can be used to create a mapping that links different vendor-provided names referring to the same actor.

Such mappings are also maintained by the community and by reputable vendors, including MISP Threat Actor Galaxy (TAG) [61]; APTMap (which uses TAG and other sources) [62]; Florian Roth’s spreadsheet [63]; Thai CERT [64]; MITRE [19]; Breach-HQ (769 actors) [65]; Qianxin [66]; and vendor-specific mappings [67], [68]. However, all mappings suffer from a lack of ground-truth. They present someone’s best understanding that two different names used by different vendors represent the same actor, while keeping up with a dynamic threat-actor landscape. Earlier work, in the context of AV, highlighted difficulties in measurement of AV performance when vendors provide different virus labels [69], [70]. Nonetheless, these mappings are widely used in academic research [20], [26], [71]–[73] and within industry [74], [75].

4. Methodology

To perform this study, we partnered with a national government enterprise organization with several thousand employees that purchases TI feeds for monitoring and analysis. The partner agreed to collaborate under two conditions: (i) anonymity of the partner organization and (ii) anonymity of the commercial TI sources they acquired. We discuss more details, implications and ethical considerations regarding this collaboration in Section 9. All feeds are sourced by the partner organization from top-tier commercial threat

TABLE 1. OVERVIEW OF ACTOR-ATTRIBUTED IOCS PER VENDOR. VENDOR ACTOR COVERAGE IS RELATIVE TO TAG’S TOTAL OF 855 ACTOR OBJECTS. PARENTHESES INDICATE WIP LABELS.

ID	Total IOCs	Unique Actors	Actors in TAG	Coverage of TAG	IOCs with Actor in TAG
V1	382,991	721 (599)	147	17%	80,361 (21%)
V2	12,633,525	229 (0)	146	17%	12,599,531 (99%)
V3	85,089	110 (0)	88	10%	54,915 (64%)
V4	384,109	105 (14)	68	8%	64,723 (16%)
V5	11,335	131 (0)	38	4%	2,616 (23%)
V6	22,312	130 (0)	89	10%	17,701 (79%)
V7	12,110	39 (0)	18	2%	7,009 (57%)

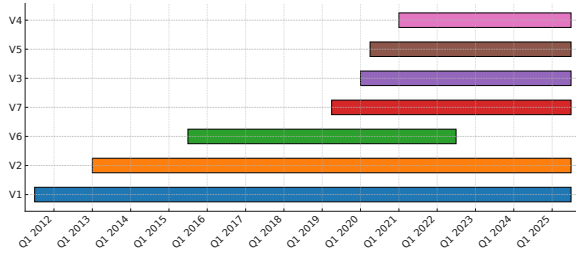


Figure 1. Overview of collection window timelines

intelligence vendors that focus on APT tracking. The partner organization subscribes to multiple top-tier commercial feeds to increase coverage. Vendor selection is driven by enterprise procurement decisions, not research cherry-picking. We used IOC data that was available on the internal systems of the partner organization, no IOCs were filtered by the research team.

We examine actor and country attribution of IOCs published by seven commercial TI vendors (designated as V1-V7), including only the records that contain an explicit actor attribution by the vendor. Table 1 shows the number of attributed IOCs per vendor. In total, our dataset comprises 13,685,963 actor-attributed IOCs spanning 14 years. However, the collection windows of IOCs per vendor vary, as described in Figure 1. For all vendors except V6, the last IOC was collected in June 2025. Note that for six vendors, we have IOCs spanning for at least five years or longer. This overlap in time means they were making attributions in the same time frame.

While we are not allowed to disclose the vendor names, we can state that five of the seven vendors appear in the MISP list of threat intelligence producers [76]; three are listed as NCSC-UK Assured Cyber Incident Response Providers [77]; and four are covered by Gartner Peer Insights for Threat Intelligence Products and Services [78]. Each vendor appears in at least one of these sources. Vendor V7 has a regional focus, whereas the others do not. Within industry, these feeds are considered top-tier, as is reflected in their pricing. The annual fees per feed (including reports) vary between \$150,000 and \$350,000.

4.1. Normalization of Heterogeneous Feeds

Vendor feeds arrive in heterogeneous formats, including flat JSON-like lists, STIX 1.x/2.x objects [79], and MISP

Event/Object data [80]. We transform each feed into a flat schema (Table 2) to enable cross-vendor comparison.

We first normalized their IOC data and put them into a single scheme. We renamed any indicator type referring to an IPv4 address to IPV4, any domain to DOMAIN, and any URL to URL. Next, the feeds were inconsistent in what hashing they applied: some vendors included multiple types (e.g., SHA1, SHA256, MD5) within a single IOC record, while others created separate records for each type. Additionally, vendors used different types of hashes. However, all vendors provided at least an MD5 hash in most cases. Therefore, we use the MD5 hash type to compare among vendors. We omitted 8,130 IOCs (0.06% of all attributed IOCs) that did not provide an MD5 hash type.

The preceding steps resulted in four indicator types (IPV4, DOMAIN, URL, MD5) that were present in every feed. We refer to those types as the *primary indicator types*. In our dataset, 13,423,585 indicators belonged to the four primary indicator types (98%). Figure 10 depicts the number of IOCs per indicator type per vendor. For the remaining 2%, we identified 55 other (secondary) indicator types. This included, for example, a YARA rule or an X509 certificate. Two feeds included combinations of primary indicator types, such as the type domain|ip. We converted an IOC record with such an indicator type into two IOC records, one with the indicator type set to DOMAIN and one with the indicator type set to IPV4, copying the other fields. We converted 6 combined-type IOCs to primary indicator types. We did not apply any transformation to the other indicator types.

4.2. Actor Name Mapping

Within this study, we use the MISP TAG mapping, specifically a snapshot from 29 July 2025 with commit 42b5d56. We use this source for several reasons. First, MISP TAG is an open project that is actively maintained [81]. Second, it provides information on the largest number of actors (Table 5 in Section A). Third, TAG is recognized and used by other popular tools and platforms. For example, Malpedia, a vetted curation platform offering a corpus of labeled, unpacked malware samples, uses TAG in mapping actors to malware [74].

However, our preliminary analysis indicates that this mapping still has room for improvement. Therefore, below

TABLE 2. NORMALIZED IOC SCHEMA.

Name	Description
id	Universally unique IOC identifier
vendor_id	IOC identifier from vendor
created	IOC creation date (UTC)
type	Indicator type (e.g., domain, ipv4)
value	Indicator value
vendor	Anonymized vendor identifier
actor	Vendor attributed actor name
actor_id	Normalized actor ID
country	Attributed country (optional)

we describe the steps taken to validate TAG and enhance the mapping using TI vendor reports.

4.2.1. Actor Objects, Names and WIP Labels. TAG uses the following JSON-like structure to describe an actor [81]:

```
[{"name": "<actor_name>",
  "synonyms": ["<alias_1>", "<alias_2>", "..."],
  "country": "<country>"
}, ...]
```

Note that `name` in TAG serves as both a name and also as the identifier. To make an explicit distinction between actor objects and actor names (the set of all associated names, including aliases/synonyms), we posit that an actor object has an ID and a list of names, which are all the names, synonyms, and aliases for the actor. Therefore, for our analyses, we adopt a simplified format to avoid ambiguity between the `name` and `synonyms` fields that exist in TAG:

```
[{"id": {
  "names": [<all_known_actor_name_and_aliases>],
  "country": <country_or_null>
}, ...]
```

In this schema, `<id>` is an auto-incremented identifier for the actor object; `names` is a deduplicated set of strings with names, aliases, and synonyms for that actor. In this study, when we refer to an *actor*, we imply an *actor object*.

Last, as described in Section 3, various vendors attribute IOCs with a type of "work in progress" or "uncategorized" label, which we dub a WIP label. Usually, an actor with a WIP label gets merged with a known actor. For example, this is the case with UNC1945, which later was linked to actor LightBasin. In reality, though, not all WIP labels get merged. In TAG, in 29 cases (out of 855, 3%), an actor has a WIP label and no synonyms. In our study, if an actor occurs in only one feed, has a WIP label, and has no synonyms, we refer to it as a WIP actor. We identify this subset so we can later look at attribution agreement with and without this set of actors.

4.2.2. Country Attribution. One vendor provides an explicit country attribution in the feed. In six out of seven feeds, a country label was not explicitly a part of the IOC feed. For four vendors, country attribution can be inferred from actor naming conventions (e.g., "Bear" for Russia, "Panda" for China). Two remaining feeds did not provide country attribution explicitly or through naming conventions. For these, we used the country of the actor as mapped in the TAG dataset; otherwise, we derive it from the corroborating vendor reports used to extend the mapping. Because not all actors are state-linked, the country can be null.

4.2.3. TAG Validation and Augmentation. Given the importance of TAG for our analysis, as well as its value for practitioners in industry, we validate and augment the TAG data. Specifically, we check for ambiguous mappings of names to actors and compare the TAG mapping against explicit claims from the vendors themselves. We also augment TAG by adding additional names and mappings from

vendor reports. To achieve this goal, at first, for each actor not mapped in TAG, we looked up the latest 3 to 5 TI reports of the vendors that mentioned that actor. Next, for those actors, we identify aliases from the TI reports. For example, a report might say "ActorX AKA ActorY", or "ActorX (ActorY, ActorZ)", or "ActorY, which we track as ActorX", or variations hereof. These examples are not exhaustive, and we encountered many small variations. For this particular example, these statements would allow us to link the names "ActorX, ActorY, and ActorZ" to the same actor with multiple names.

4.3. Agreement Metrics and Analysis

Our goal is to quantify cross-vendor agreement on actor attribution for the same IOCs. We define co-observation of an IOC across vendors as an exact match on the normalized *indicator value* field – e.g., `evildomain.com` or `192.0.2.34`.

However, we should also take a time dimension into account because IOCs have a limited lifespan. If an IOC is observed once by one vendor and once by another, but those observations are far apart in time, then they are less likely to have been part of the same threat. So when those vendors attribute them to different actors, that might not signal disagreement about the attribution but simply mean that the threats belong to different actors. For this reason, we want to look at agreement for co-observed IOCs where the observations by different vendors are temporally close to each other.

How close? There is prior work on IOC lifetimes [35]–[38], but no consensus on estimates of IOC lifetime. Therefore, we analyze our data to see how different lifetimes affect the agreement levels. Given the fleeting nature of IOCs, we introduce several observation periods (OPs): 7, 14, 30, and 60 days. For example, for OP=7, we analyze IOCs that were observed by two or more vendors within a period of 7 days. This way, we can see if the agreement level is sensitive to the temporal distance between the observations of the same IOC.

For the co-observed IOCs, we compute inter-rater reliability (IRR) statistics on nominal labels to determine the level of agreement among observers. We treat the attribution of co-observed IOCs as a classification task, where each vendor assigns an actor label to an IOC. We measure agreement using Krippendorff's alpha [82], which accommodates multiple raters and nominal data. It takes chance agreement into account, penalizing any kind of disagreement. Krippendorff's alpha ranges from -1 (perfect systematic disagreement) to 1 (perfect agreement). To qualify the level of agreement, we interpret Krippendorff's alpha as it is conventionally understood in the context of IRR [83]: $\alpha < 0$ indicates systematic disagreement, $\alpha = 0$ indicates chance-level agreement, $0 < \alpha < 0.67$ indicates poor agreement, $0.67 \leq \alpha \leq 0.79$ indicates moderate agreement, $\alpha \geq 0.80$ indicates satisfactory agreement, and $\alpha = 1$ indicates perfect agreement. We compute IRR statistics for sets of two, three,

or four vendors co-observing IOCs, as no IOCs are co-observed by five or more vendors. We estimate 95% confidence intervals (CI) for α using a nonparametric bootstrap with 500 resamples, to determine the reliability of the α values.

Furthermore, to determine the agreement among specific vendors, rather than among any pair of vendors, we also calculate pairwise agreement. For each vendor pair and period, we include IOCs for which both vendors assigned an actor label, count matches versus mismatches (i.e., assign the same actor or country), and report the proportion of matches. This measure is a percent agreement in $[0, 1]$, not chance-corrected (unlike Krippendorff’s α), where 1 indicates perfect agreement.

Finally, in Section 7.2, to get a better understanding of observed disagreement, we use VirusTotal (VT) [84] and look up 867 hashes to determine the *file type* and *malware label*.

5. TAG Validation and Augmentation

An accurate mapping of vendor actor names is vital to compare attributed IOCs. So as we use TAG in this study, we validate its quality by i) checking TAG for ambiguity in actor names, and ii) measuring the validity of the mapping against claims of the vendors themselves.

5.1. Actor Ambiguity in TAG

TAG defines mappings between actor objects and their associated names (aliases or synonyms). In total, TAG contains 2,080 distinct names linked to 855 actor objects. However, these mappings are not always one-to-one: some names are associated with multiple actor objects, introducing ambiguity in the name-to-actor relationship.

For example, the name Evasive Panda appears both as the primary name of one actor object and as an alias for another actor, BRONZE HIGHLAND. As a result, Evasive Panda cannot be treated as a unique identifier for a single actor. A naïve approach might be to merge actor objects that share a name, assuming they refer to the same entity. Yet, such merges can propagate inconsistencies, as, once merged, the combined alias set may overlap with additional actor objects, causing cascading ambiguities that obscure the underlying actor structure.

We refer to this phenomenon as “alias ambiguity” (AA). In total, from the 2,080 actor names in TAG, we found 23 such ambiguous names (1%), presented in Table 8 in Section A. We excluded the associated 22,477 IOCs (0.2% of all IOCs).

5.2. Validating TAG

We want to test whether TAG accurately reflects the claims of the vendors themselves when it comes to mapping different vendor names to the same actor. In other words, do the vendors concur that these mappings are accurately

reflecting their understanding? We perform two validation steps to check the accuracy of TAG in representing the vendor’s own statements about a shared attribution. First, in June 2025, CrowdStrike and Microsoft announced a bilateral reconciliation of actor names, publishing a cross-vendor mapping (CS/MS Mapping) [85]. We compare TAG against this mapping. Second, we randomly sample 50 actor names from TAG and check TAG mappings against the claims of the vendors in their reports.

5.2.1. CrowdStrike and Microsoft Mapping. The CS/MS Mapping published by CrowdStrike (CS) and Microsoft (MS) lists 84 actor mappings, where a name used by CrowdStrike is matched against the corresponding name of Microsoft. We matched each CS/MS name pair to TAG and asked whether both names resolve to the same TAG actor – either as the primary TAG name or within the synonyms. Overall, 83% (70/84) of pairs were present and consistent with TAG. Four pairs, so eight distinct names, were absent from TAG (i.e., item A in Table 9). We added these to our augmented TAG.

Eight pairs were present in TAG but resolved to different TAG actors (i.e., item B in Table 9), i.e., the CS/MS Mapping treats each pair as names of a single actor, whereas TAG treats them as names of two distinct actors.

For example, in TAG, Ruby Sleet is described as an actor linked to North Korea’s Ministry of State Security, and Kimsuky as a North Korean actor that targets South Korean think tanks, industry, nuclear power operators, and the Ministry of Unification for espionage purposes. These actors could potentially be the same actor or actors in different divisions within the same organization, the North Korean Ministry of State Security.

We consider the second case (item B) similar to the “actor ambiguity” described in Section 5.1, and therefore exclude IOCs with those actor names. This affects 103,815 IOCs (0.9% of all IOCs).

5.2.2. Validating Against TI Reports. To verify the quality of TAG mappings, we randomly sampled 50 actors and their names from TAG. For each sampled actor and its names (aliases), we examined the latest TI reports that mentioned the aliases to extract explicit alias statements. Such statements are, for example, when a vendor name is followed by “AKA” and a list with names of other vendors. This indicates that the vendor itself sees these as the same actor. Another example is the phrase “we track X as Y”. We compared all such statements with the TAG entries. We observed no discrepancies. In all 50 cases, the mappings that the reports explicitly contained, were all consistent with TAG.

5.3. TAG Augmentation

We examined how many of the actors from a vendor feed are listed in TAG. The number of TAG-listed actors per vendor is depicted in Table 1. For example, 147 (of all 721 actors tracked by V1) are listed in TAG, with 80,361

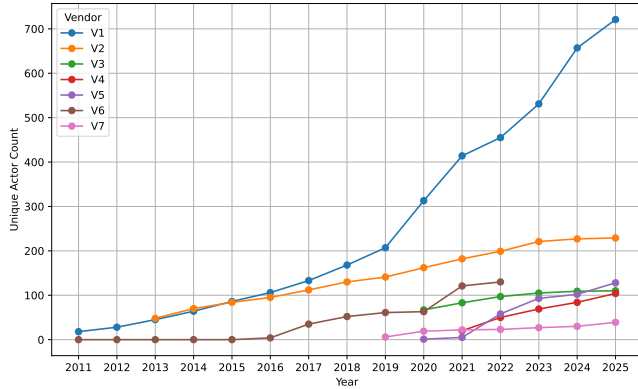


Figure 2. Cumulative number of tracked actors per vendor over time, includes WIP actors.

(21% of all attributed IOCs by this vendor) IOCs attributed to those actors.

The results show that all vendors have a portion of their actors not showing up in TAG. Thus, we aimed to augment TAG. We do so by examining vendor TI reports for known aliases for those actors that occur in the vendor feeds but are not listed in TAG, using the format described in Section 4.2.1. The result of our augmentation is a new mapping that consists of the original TAG mappings, plus additional actors and aliases.

5.3.1. Ambiguity in Mapping. In our augmentation process, we ran into ambiguous mappings done by vendors. For example, a vendor stating ‘ActorX (AKA ActorY, ActorZ, ActorC)’ may be problematic when ActorY and ActorZ are associated with distinct actor objects in TAG. In those cases, mapping ActorX would be ambiguous because we do not know whether we should map it with actor object ActorY or actor object ActorZ in TAG. This is similar to the “actor ambiguity” described in Section 4.2.

To not add ambiguity to TAG, we did not include those mappings. For our subsequent analyses, we also filtered out the associated IOCs. We encountered this situation 11 times, excluding 6,701 IOCs (0.05%).

5.3.2. Augmentation Results. We augmented the TAG mapping with 871 unique actors and 4 countries, nearly doubling the number of actors in TAG. The total number of actors in the augmented TAG is 1,726, and the total number of countries is 41. However, out of the 871 actors added, 833 actors appear in only one vendor feed; they are singletons. Furthermore, 614 of the added actors have a WIP label as a name (e.g., DEV-XXX or TAG-XXX, see Section 4.2.1). In sum, the majority of the augmented mapping consists of actors with no aliases, and frequently with a WIP label.

6. Results: Actor Tracking

We now address RQ1: *what is the scope of actor-tracking by vendors?* We operationalize the scope of actor-tracking along several dimensions. First, we analyze the

number of actors reported by vendors over time to understand trends in actor-tracking growth and the relative coverage of actors per vendor. We also aim to estimate how much of the overall actor population is reported by vendors. Since no ground truth exists for the complete population, we use TAG as a proxy for the broader actor landscape tracked by entities within the TI community, as reflected by their inclusion in TAG. Next, we examine the overlap in tracked actors between vendors and assess how many additional actors are captured when a new vendor is added. Finally, we focus on actor-specific tracking by pairs of vendors: when two vendors track the same actors, do their IOCs overlap, or does each vendor report different artifacts associated with the actor?

6.1. Tracked Actors Over Time

For each vendor and calendar year, we identify actors first observed in that year (i.e., not seen in earlier years for that vendor). We then compute the cumulative sum of these uniquely observed actors over time. These counts are limited to the periods for which we had access to the IOC feeds (see Figure 1). Vendors may have published outside our collection window, so the true number of tracked actors is likely higher.

In Figure 2, all vendors show steady growth in the number of tracked actors. One vendor, V1, sharply increases the number of tracked actors from 2019 onward. This is primarily due to the use of WIP labels. Our feed for vendor V6 ends in 2022. Table 1 summarizes totals per vendor; V1 has the highest number of tracked actors (721), V7 has the lowest (39). However, the totals also reflect the time periods during which the feeds were collected. Outside those periods, additional actors are likely covered.

Across all seven vendors, we identify 1,155 unique actors (565 excluding WIP labels). Table 1 depicts the actors per vendor and the number of actors in TAG that are reported on. It shows that not a single vendor reports on more than a fifth of all threat actors listed in TAG. Individual vendors vary greatly in reporting on the number of actors listed in TAG, with V7 reporting on only 2% of all TAG actors, and V1 and V2 reporting on 17%. If we combine the tracked actors of all vendors, then together they report on 293 actors listed in TAG, 34% of all TAG actors (855). In other words, an organization that purchases all seven IOCs feeds, at a cost of USD 1-2 million per year, would still capture only 34% of the threat actor landscape as identified by the industry as a whole.

6.2. Overlap and Union in Tracked Actors

We determine the overlap and union of tracked actors among vendors. In doing so, we compare the pairwise overlap of tracked actors for two vendors and the union of tracked actors between multiple vendors.

First, we want to know to what extent vendors track the same actors. To do so, we calculate the pairwise overlap of tracked actors among vendors using the

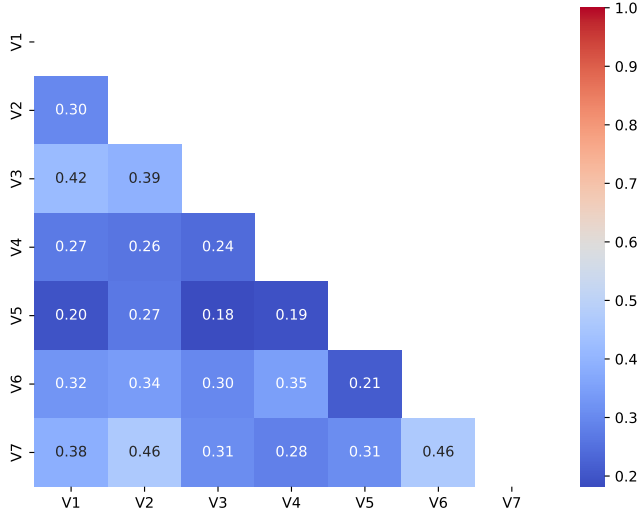


Figure 3. Overlap coefficient ($\mathcal{O}$) of tracked actors between vendors.

Szymkiewicz–Simpson overlap coefficient $\mathcal{O}$. We use the overlap coefficient because the sizes of the participating sets (i.e., tracked actors per vendor) are very different, and the overlap coefficient normalizes to the smaller set. Let A_v denote the set of actors reported by vendor v under our augmented TAG mapping (where an actor is associated with a list of names). For each vendor pair (i, j) we compute $\mathcal{O}$, defined as:

$$\mathcal{O}(i, j) = \frac{|A_i \cap A_j|}{\min\{|A_i|, |A_j|\}}.$$

Figure 3 shows a heatmap of $\mathcal{O}(i, j)$ across all vendor pairs (we also include Venn diagrams in Figure 8 in Section A.) Across all 21 vendor pairs, the mean overlap coefficient is 0.31, indicating modest overlap. The minimum overlap coefficient is 0.18, and the maximum overlap coefficient is 0.46. Thus, for all pairs, the majority of actors tracked by one vendor are not tracked by the other, and each vendor seems to focus on a mostly different part of the actor landscape.

Second, how much does each additional vendor add in terms of coverage of the actor landscape? In Figure 4, we show the number of new actors each new vendor can add to the total set. For each additional vendor, we select the vendor that adds the highest number of additional actors. We show this for actors included in the original TAG mapping and for actors we added to TAG in our augmentation. We see that in both cases, there are diminishing returns. The seven vendors combined cover 293 actors from the original TAG mapping, which is shown in Figure 4. This covers 34% of all original TAG actors. The union of augmented TAG actors follows a similar pattern, but does not diminish at the same rate, reaching over 500 unique actors in total.

A 2018 study found that organizations used approximately 6 to 10 different TI feeds (of which 3-5 are commercial sources) [86]. Thus, most organizations use attributed TI that covers only a minor part of the threat actor landscape.

Given the hefty price tags on commercial TI, getting TI that reports on most (TAG) actors would be prohibitively expensive, given that even our seven vendor feeds cover only a third of the TAG actors.

6.3. IOC Overlap for Jointly Tracked Actors

In Section 6.2, we observed that the overlap in tracked actors is limited. We now turn to the IOC level to assess how comprehensive the visibility of two vendors is when tracking the same actors. Specifically, to what extent do the vendors’ IOCs for these actors overlap? As in the previous analysis, we use the overlap coefficient to account for size imbalances between vendors’ IOC sets.

For each vendor pair (i, j) and each actor a present in both A_i and A_j , we form the sets of unique IOCs $I_{i,a}$ and $I_{j,a}$. We compute $\mathcal{O}(I_{i,a}, I_{j,a})$ by collecting all IOCs per vendor for those actors and computing the intersection of those IOCs between vendors. We use these inputs to compute the overlap coefficient for the pairwise vendor matrix in Figure 5.

The average of all pairwise overlap coefficients is 0.10. This shows that across vendors there is little overlap of IOCs. Note that this is calculated only including the IOCs of those actors that the vendors claim to be tracking. Even when vendors track the *same* actor, the IOCs they publish barely overlap. In other words, the visibility of each vendor on an actor is very limited. One potential explanation is the different kinds of telemetry that vendors have, which would be reflected in the indicator types. Figure 10 in Section A shows the number of IOCs per indicator type for each vendor. It is evident that vendors vary in the types of indicators they release. Several vendors provide MD5 hashes most frequently, followed by domains. Only one vendor predominantly provides IPv4 indicators.

These findings align with prior work showing limited shared observables across open [28] and commercial TI sources [26]. The latter study only analyzed the overlap among two vendors that jointly tracked 22 actors. It reported

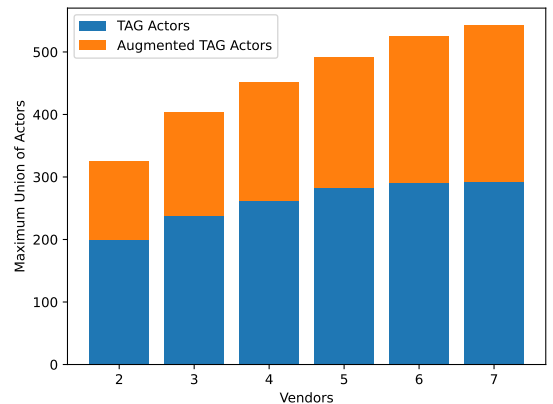


Figure 4. Maximizing actor union between vendors overall and in TAG, excluding WIP actors.

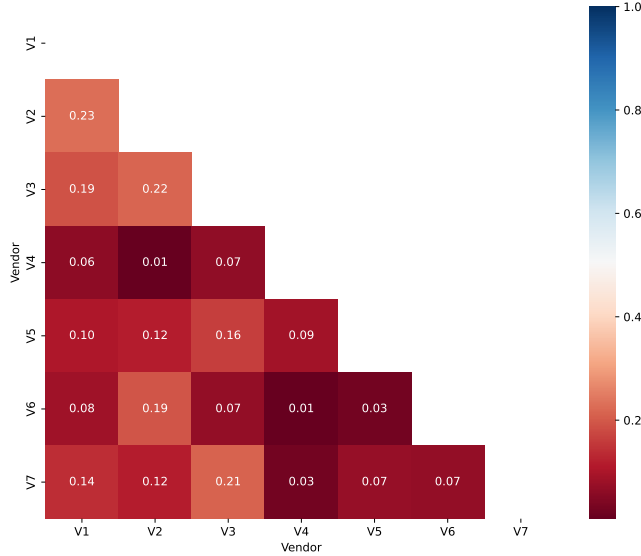


Figure 5. Overlap coefficient (O) for IOCs associated with actors that are tracked by both vendors.

between 2.5% and 4% overlap for two vendors. In our study, based on a more comprehensive dataset, we do find a slightly larger overlap, namely of 10% on average.

Across all pairwise vendor comparisons and indicator types, we observe an average overlap of approximately 10% – i.e., an overlap coefficient of 0.10 – with a minimum of 1% and a maximum of 23%, indicating substantial variability in overlap among vendors. While V1 and V2 have high overlap, they are also the vendors with the longest collection window, depicted in Figure 1. However, V3, with a much shorter collection window, also has high overlap with several vendors. This suggests that higher overlap is not merely due to longer collection windows.

7. Results: Attribution Agreement

We now turn to our second research question (RQ2): *How much agreement is there on the attribution of actors and countries across vendors?* We answer this question by looking at IOCs that are observed and attributed by two or more vendors. For example, the domain `evildomain.com` may be reported in two IOC feeds. One vendor may attribute it to ActorA while another vendor to ActorB. For simplicity, we will refer to these as ‘co-observed IOCs’. However, note that ‘co-observed’ also means that both vendors attributed the IOC, possibly to different actors. At the same time, most IOCs in the feeds are not attributed by anyone or only by a single vendor. Those are excluded from this analysis. As described in Section 4.3, we measure the agreement in attribution using the Krippendorff’s α metric for IRR. For each co-observed IOC, we only consider the vendors that ‘voted’ (i.e., attributed) the actor. So there are no ‘missing votes’. Using this approach, though, we cannot compare specific vendors, only overall cross-vendor agreement.

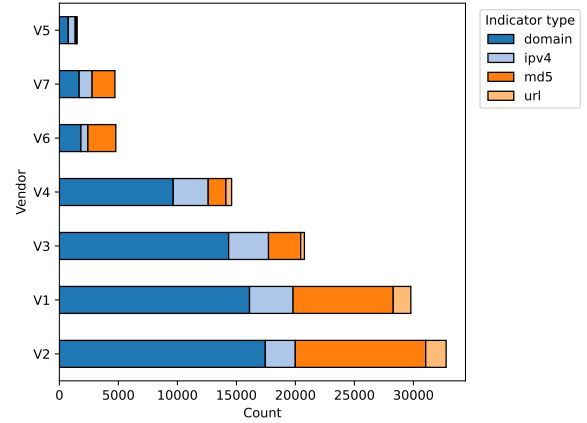


Figure 6. Distribution of actor attributed IOCs co-observed by 2+ vendors, excluding WIP labels.

As discussed in Section 4, we use observation periods (OP) to ensure that the different observations of the same IOC happen within a certain timeframe. For example, for $OP=7$, we analyze IOCs that were observed by two or more vendors within a period of 7 days. If the observations are very far apart, they are less likely to belong to the same attacker and, thus, less likely to be both correctly attributable to that actor. So this would bias the agreement metrics against the vendors. We want to be conservative, so we only compare IOCs that are closer in time. To understand if the observation period impact the agreement, we analyze our data for different periods: 7, 14, 30, and 60 days. In Section 7.1, we describe the number of co-observed IOC overall and per observation period. First, we examine agreement over indicators for co-observed IOCs that include WIP label attributions (described in Section 4.2.1). However, we do not consider WIP attributions to be real *actor* attribution. Additionally, they lead to disagreement because the label is vendor-specific. Therefore, the subsequent analyses consider attributions excluding WIP labels. Next, we investigate the set of co-observed IOCs by two and three vendors. The number of observations by four vendors (see Table 10) was so low that we cannot derive any conclusions. No IOCs were co-observed by five or more vendors. Furthermore, in Section 7.2, we try to understand disagreement for MD5 hashes by looking at the types of software, obtained from VirusTotal (VT) [84]. In Section 7.3, we determine how specific vendors relate to each other when co-observing IOCs by computing the pairwise agreement between specific vendors for different observation periods. This analysis does not include IOCs with WIP labels.

7.1. Co-observed IOCs

IOCs that are co-observed and attributed by two or more vendors are rare. If we do not apply any limitation on how far apart the observations took place, then 135,987 IOCs (1% of all attributed IOCs, including WIP labels) are co-observed. Excluding WIP labels, there are 112,625 IOCs.

TABLE 3. ACTOR AND COUNTRY ATTRIBUTION AGREEMENT BY INDICATOR TYPE AND OBSERVATION PERIODS (OP). VALUES SHOW KRIPPENDORFF’S α WITH 95% CI AND SAMPLE SIZE (n). ALL TYPES EXCLUDE WIP LABEL IOCS, EXCEPT ALL (INC.WIP).

Type	OP (Days)	2 vendors		3 vendors	
		Actor	Country	Actor	Country
ALL (inc. WIP)	7	0.60 [0.59,0.61] ($n=3,530$)	0.89 [0.88,0.91] ($n=2,761$)	0.50 [0.42,0.60] ($n=47$)	0.64 [0.58,0.72] ($n=44$)
ALL (inc. WIP)	14	0.61 [0.60,0.62] ($n=5,599$)	0.92 [0.91,0.93] ($n=4,612$)	0.68 [0.63,0.73] ($n=144$)	0.89 [0.85,0.95] ($n=129$)
ALL (inc. WIP)	30	0.56 [0.55,0.57] ($n=10,089$)	0.91 [0.90,0.92] ($n=7,729$)	0.68 [0.64,0.71] ($n=344$)	0.81 [0.76,0.86] ($n=322$)
ALL (inc. WIP)	60	0.57 [0.56,0.57] ($n=15,822$)	0.89 [0.89,0.90] ($n=12,519$)	0.67 [0.65,0.69] ($n=851$)	0.85 [0.82,0.88] ($n=777$)
ALL	7	0.81 [0.80,0.82] ($n=2,748$)	0.94 [0.93,0.95] ($n=2,487$)	0.71 [0.49,1.00] ($n=35$)	1.00 [1.00,1.00] ($n=35$)
ALL	14	0.79 [0.78,0.80] ($n=4,524$)	0.95 [0.94,0.96] ($n=4,201$)	0.83 [0.76,0.92] ($n=120$)	1.00 [1.00,1.00] ($n=120$)
ALL	30	0.78 [0.77,0.79] ($n=7,642$)	0.95 [0.94,0.95] ($n=7,060$)	0.81 [0.76,0.86] ($n=295$)	0.97 [0.95,1.00] ($n=295$)
ALL	60	0.76 [0.76,0.77] ($n=12,326$)	0.92 [1.00,1.00] ($n=11,428$)	0.76 [0.74,0.79] ($n=730$)	0.95 [1.00,1.00] ($n=729$)
MD5	7	0.83 [0.81,0.85] ($n=867$)	0.97 [0.96,0.99] ($n=695$)	1.00 [1.00,1.00] ($n=11$)	1.00 [1.00,1.00] ($n=11$)
MD5	14	0.83 [0.82,0.85] ($n=1,449$)	0.98 [0.97,0.99] ($n=1,234$)	1.00 [1.00,1.00] ($n=55$)	1.00 [1.00,1.00] ($n=55$)
MD5	30	0.84 [0.82,0.85] ($n=2,486$)	0.98 [0.97,0.98] ($n=2,158$)	0.99 [0.98,1.00] ($n=132$)	1.00 [1.00,1.00] ($n=132$)
MD5	60	0.83 [0.82,0.84] ($n=3,753$)	0.96 [0.95,0.96] ($n=3,282$)	0.90 [0.85,0.95] ($n=314$)	0.90 [0.85,0.95] ($n=314$)
URL	7	0.87 [0.83,0.90] ($n=275$)	0.99 [0.98,1.00] ($n=243$)	1.00 [1.00,1.00] ($n=1$)	1.00 [1.00,1.00] ($n=1$)
URL	14	0.89 [0.86,0.91] ($n=407$)	0.99 [0.98,1.00] ($n=365$)	1.00 [1.00,1.00] ($n=1$)	1.00 [1.00,1.00] ($n=1$)
URL	30	0.82 [0.80,0.85] ($n=625$)	0.99 [0.98,1.00] ($n=556$)	1.00 [1.00,1.00] ($n=9$)	1.00 [1.00,1.00] ($n=9$)
URL	60	0.87 [0.85,0.89] ($n=951$)	0.99 [0.98,1.00] ($n=844$)	0.70 [0.61,0.79] ($n=44$)	1.00 [1.00,1.00] ($n=44$)
DOMAIN	7	0.84 [0.83,0.87] ($n=1,236$)	0.98 [0.97,0.99] ($n=1,189$)	0.48 [0.45,0.49] ($n=22$)	1.00 [1.00,1.00] ($n=22$)
DOMAIN	14	0.78 [0.77,0.80] ($n=1,916$)	0.99 [0.98,1.00] ($n=1,864$)	0.66 [0.55,0.78] ($n=59$)	1.00 [1.00,1.00] ($n=59$)
DOMAIN	30	0.79 [0.78,0.80] ($n=3,224$)	0.98 [0.97,0.99] ($n=3,103$)	0.63 [0.56,0.70] ($n=138$)	0.94 [0.86,1.00] ($n=138$)
DOMAIN	60	0.76 [0.75,0.77] ($n=5,554$)	0.94 [0.93,0.95] ($n=5,351$)	0.65 [0.61,0.70] ($n=294$)	0.94 [0.88,1.00] ($n=293$)
IPv4	7	0.65 [0.61,0.69] ($n=278$)	0.93 [0.90,0.96] ($n=268$)	0.00 [0.00,0.00] ($n=1$)	1.00 [1.00,1.00] ($n=1$)
IPv4	14	0.71 [0.68,0.74] ($n=601$)	0.96 [0.95,0.98] ($n=587$)	0.78 [0.55,1.00] ($n=5$)	1.00 [1.00,1.00] ($n=5$)
IPv4	30	0.66 [0.63,0.68] ($n=1,070$)	0.96 [0.95,0.97] ($n=1,008$)	0.81 [0.68,1.00] ($n=16$)	0.93 [0.83,1.00] ($n=16$)
IPv4	60	0.66 [0.64,0.68] ($n=1,698$)	0.94 [0.93,0.95] ($n=1,584$)	0.65 [0.59,0.70] ($n=78$)	0.96 [0.93,1.00] ($n=78$)

The distribution of co-observed indicator types per vendor is depicted in Figure 6. It shows that hashes and domains are most frequently co-observed, URLs the least. Also, vendors V1 and V2 account for a large portion of co-observed IOCs.

As soon as we select an observation period, the maximum distance in time between the observations of different vendors is set to 7, 14, 30, or 60 days. The smaller the OP, the lower the number of co-observed IOCs. Table 3 shows the number of co-observed IOCs for each observation period (OP) for actor and country attribution. As expected, the most common case is two vendors co-observing an IOC within the 60-day OP, namely 12,326 IOCs (15,822 exc. WIP labels).

For the country attribution, there are 11,428 IOCs (12,519 exc. WIP labels) for two vendors over a 60-day OP. The maximum number of vendors that co-observe an IOC is four, depicted in Table 10. No IOCs are co-observed by five or more vendors.

7.1.1. Co-observations of IOCs including WIP labels.

Table 3 reports the overall agreement for IOCs including WIP labels (type=ALL (INC.WIP)). We find poor agreement across observation periods: α ranges from 0.56-0.61. Agreement among vendors is the highest when the observation period is shorter ($OP \leq 14$), and declines a bit ($\alpha = .56$ and $\alpha = .57$) when the observation period is increased to 30 and 60 days. This decrease, however, is small, so the chosen observation period does not seem to have a substantial impact on the level of agreement. By contrast, country attribution has high agreement across all observation periods and indicator types (α ranges

from 0.89-0.92). These observations are similar for co-observed IOCs by 3 vendors. Vendors are, thus, likely to disagree over actors but not over the origins of the attacks.

As stated before, we do not consider WIP label IOCs strong attributions, and neither do vendors [58]. We focus all analysis below on agreement for attributions *excluding* WIP labels.

7.1.2. Co-observation by two vendors. Table 3 shows that IOCs co-observed by two vendors are most prevalent. For type ALL, we find moderate agreement across observation periods: α ranges from 0.76-0.81. Agreement among vendors is highest when the observation period is shortest ($OP \leq 7$), and marginally declines ($\alpha = .78$ and $\alpha = .76$) when the observation period is increased to 30 and 60 days. This is not surprising, as attacker infrastructure more likely to have been abandoned and changed hands over time [38]. The chosen observation period does not seem to have a substantial impact on the level of agreement. Conversely, country attribution approaches near-perfect agreement across observation periods and indicator types (α ranges from 0.92-0.95).

Agreement among vendors is lowest for IPv4 IOCs (α ranges from 0.65-0.71). Also, IPv4 indicators are co-observed less often. The agreement is moderate (α ranges from 0.65-0.71) and remains somewhat constant across observation periods. This relative disagreement is perhaps not surprising as IPv4 addresses are believed to be transient attacker infrastructure with high churn [17], [38].

URL indicators are the least observed IOCs. Similar

to MD5 indicators, they have a satisfactory and stable level of agreement for all observation periods ($\alpha > 0.80$). Contrary to IPv4 addresses, URLs and hashes tend to be of a more permanent nature, which seems reflected in the level of agreement. For this reason, operational security (OPSEC) best practices dictate that files and payloads should not be reused [87]

DOMAIN indicators have the least stable agreement across observation periods. These IOCs have high agreement at the shortest observation period, which consistently declines (α ranges from 0.84-0.76). This may be due to infrastructure churn when domains are identified as malicious.

Overall, we find that the overall level of agreement is moderate and doesn't seem to be substantially impacted by a longer observation period. This finding somewhat undermines the idea that organizations can use attributed IOCs to detect and respond to particular adversaries effectively, particularly for IPv4 indicators. While agreement on the actor is not poor, only agreement on the country consistently scores high.

7.1.3. Three-vendor co-observation. A third vendor also observing an IOC within the same observation period is rare. In total, there are 730 cases at most (see row (ALL), OP=60, column 3 vendors). The number of co-observed IOCs is particularly low for observation periods of 7 and 14 days across the various indicator types. For country attribution, the number of IOCs is equally low, but the α values remain high compared to the two-vendor agreement.

When analyzing the alpha by indicator type, we find that the DOMAIN indicator results in poor agreement, despite several observations. By contrast, the MD5 indicator has very high agreement. This may be due to the permanent nature of hashes. The IPv4 indicator has better agreement for observation periods of 14 and 30 days (α ranges from 0.78-0.81), but the number of observations is low. Once the observations increase (OP=60), the agreement is poor. This is similar for the URL indicator, where the longest observation period yields poor agreement.

The low number of observations seems to result in less stable alphas. While the agreement is higher, the confidence intervals are wider. So the net effect for observations by three vendors is not higher agreement. For actor attribution, this undermines the idea that having multiple TI sources would make attribution easier for customers. By contrast, country attribution consistently results in high levels of agreement, regardless of the number of vendors.

7.1.4. Attribution Analysis of Secondary IOCs. Secondary IOCs include types such as Yara and Snort rules, filenames, MIME types, emails, X.509 certificate details, auth hashes, crypto addresses, passwords, user agents, and imphashes (55 in total). These are largely vendor-specific. We observe only 3,706 secondary IOCs co-observed by two vendors (3% of all co-observed *primary* indicators), with none observed by more than two vendors. All were exclusively shared between V3 and V4, likely reflecting their

vendor-specific nature. Table 11 in Section A summarizes agreement levels.

We find lower sample sizes ($n = 93-375$), low actor attribution agreement ($\alpha = 0.29-0.40$), and wider confidence intervals. Country attribution also has low n but perfect agreement ($\alpha = 1$). These results suggest that vendor-specific IOCs lead to lower actor attribution agreement. As discussed in Section 6.3, we hypothesize that such IOCs stem from vendor-specific telemetry (e.g., customer base and network vantage points), potentially introducing attribution biases.

7.2. Disagreement on MD5 Indicators

What are the sources of disagreement between the vendors? This cannot be answered without inside knowledge of not just one vendor, but all of them. This is not available to anyone, not even the vendors themselves. Moreover, in the absence of ground truth, the disagreement also cannot be traced by to ineffective practices of one vendor over another.

So we can only speculate about the sources of disagreement. For most indicator types, it is not difficult to understand why vendors might reach different conclusions. Indicators like DOMAIN, URL, IPv4 have transient properties. IP addresses are re-assigned regularly, domains might be compromised assets that change hands again later. It is more surprising that also file hashes lead to different attributions. The hashes point unambiguously at very specific artefacts that are frozen in time. Why would vendors not have higher agreement rates there than for the more transient indicators?

Using VirusTotal (VT), we took a closer look at some of the files associated with the MD5 hashes are permanent and unambiguous. We looked up 867 co-attributed hashes for a 7-day observation period for two vendors. For this set, 746 hashes resulted in agreement, 121 in disagreement. For the 746 agreement hashes, VT returned 532 results. For the 121 disagreement hashes, VT returned 68 results. In total, VT had results for 69% of the hashes.

In Table 4, we depict the counts for agreement and disagreement for several file types. In our view, there are two possible explanations for these discrepancies. First, some vendors may mislabel certain files. This is problematic because if a company relies on the corresponding feed, it could lead to false positives. Second, some files may have a dual-use nature: they can be used for benign purposes under certain conditions, but for malicious purposes under others. For example, various studies looked at how attackers are 'living off the land' and appropriating legitimate resources for malicious activities [88], [89].

Other file types, like Microsoft Excel spreadsheets or Office documents, are more likely associated with initial access attempts. They might contain exploit code that can be correlated with artefacts from other attacks. Yet they still reveal less about the attackers than the full malware packages they rely on later in a multi-stage attack. So it provides less evidence on which to base attributions. Even though the object itself is stable and unambiguous, the attribution process still operates under significant uncertainty.

TABLE 4. FILETYPES FOR WHICH VENDORS BOTH AGREED AND DISAGREED IN ATTRIBUTION

File type	Agree	Disagree	Ratio (%)
MS Excel Spreadsheet	6	11	183
Outlook	1	1	100
Win32 DLL	38	22	58
ISO image	3	1	33
Windows shortcut	61	10	16
Win32 EXE	84	15	18
VBA	9	2	22
Powershell	9	1	11
HTML	109	3	3
Office Open XML Document	44	1	2

7.3. Pairwise Agreement Among Vendors

The previous subsections examined overall attribution differences among vendors, i.e., they were not vendor-specific. We want to know if overall disagreement perhaps arises from specific vendors that mostly disagree with others. Therefore, we analyze agreement differences among specific vendors by computing the pairwise agreement, described in Section 4.3. For the pairwise agreement, we compare matches (i.e., the same attributions) against all co-observed IOCs, excluding IOCs with WIP labels.

Figure 7 shows pairwise agreement among vendors for co-observed IOCs across multiple observation windows. We find that some vendors agree more than others across observation periods, but generally speaking, disagreement is distributed across the vendor population. In some periods, V2, V3, and V6 form a consistently high-agreement cluster, whereas V1 and V5 show comparatively lower agreement with others. V7 frequently behaves as an outlier, diverging from the remaining vendors; a potential explanation is its regional focus. Agreement generally weakens as the ob-

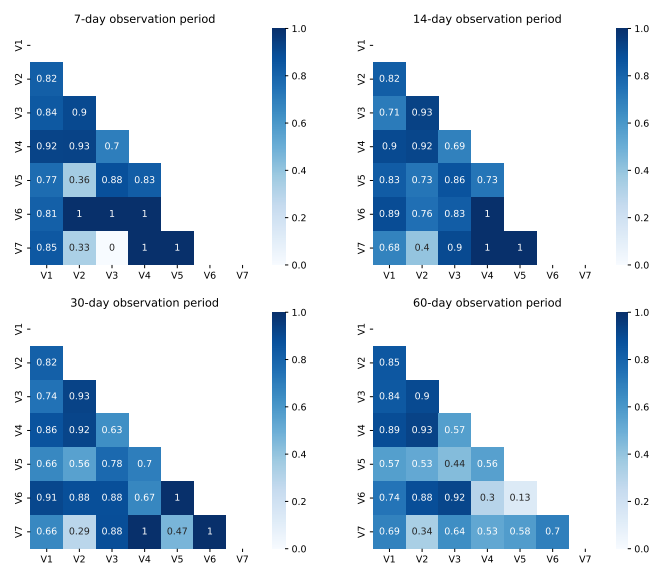


Figure 7. Pairwise vendor agreement on actor attribution for co-observed IOCs.

servation window widens, and by 60 days, no vendor pair exhibits perfect agreement.

8. Discussion

We find that vendor coverage of tracked threat actors is limited, with little overlap in attributed IOCs across vendors, even when they track the same actors. Prior work has similarly highlighted challenges in coverage, overlap, timeliness, and accuracy [23]–[26], [29].

The low overlap suggests that visibility into attacker activity is largely shaped by vendor-specific telemetry, influenced by factors such as customer base and network vantage points. As a result, each vendor captures only a partial view of the attack landscape. Whether combining these partial views yields a near-complete picture for defenders, though, remains an open question.

We now reflect on the implications of our findings for the three use cases of attributed TI, as outlined in the Introduction.

8.1. Implications

Threat-led defense strategies. For defenders, the customers of high-end TI, our findings help reduce the information asymmetry they face vis-à-vis vendors. It is very difficult for a customer to evaluate the properties and quality of TI. For the issue of attribution, we now have at least an empirical assessment of what the market delivers.

The fragmented and inconsistent nature of commercial threat attribution poses a direct challenge to threat-led defense strategies. Our findings show that each vendor captures only a fragment of the actor landscape as represented by MISP TAG. This sparsity means organizations relying on one source will have blind spots in their defenses, potentially missing adversaries that other feeds identify. Even combining all seven top-tier feeds only covered roughly one-third of known actors, indicating that defenders cannot assume coverage of the full threat spectrum. In practice, this undermines the common guidance that enterprises should ‘know the adversaries targeting them’ and might lead to misallocated investments or gaps in monitoring.

Moving to the IOCs, we found that most of them are not attributed to any actor. Where there is attribution, it is mostly by a single vendor. Even when vendors claim to track the same actors, they find mostly different IOCs. When we look at the set of attributed IOCs that are reported by more than one vendor within the same observation period, we find only modest agreement in the attribution.

Still, this is the most generous comparison we can make for vendors, and some level of disagreement still remains. This also implies that the singleton attributions, which make up the bulk of the attributed data, have a decent probability of being wrong.

Legal Exposure and Sanctions Screening. Attribution inconsistency can have serious legal and compliance ramifications. Many organizations today use threat intelligence to check whether a breach or a ransomware attack involves

actors under legal sanctions or other restrictions. For example, the U.S. Office of Foreign Assets Control (OFAC) maintains sanctions against certain cyber adversaries, and companies must avoid transactions (like ransom payments) that benefit those groups. If threat feeds provide sparse or disjoint coverage, a victim organization might fail to recognize that an attacker is a sanctioned entity simply because their intel vendor labels the group differently or not at all.

The high agreement on country-level attribution (Kripendorff's $\alpha > 0.9$) offers some consolation – at least geographically, threat intelligence largely concurs. This consistency means that if an organization knows, for instance, that any state-sponsored attacker from Country X must trigger certain legal protocols (government notifications, export control checks, etc.), the country tag from a feed is likely reliable. However, most legal obligations (like sanctions) are tied to specific named groups or individuals, not just countries.

Geopolitical Attribution and State Response. In the context of national security and state-level responses, our findings have dual implications: they highlight a challenge at the actor granularity but a reassuring alignment at the country level. Governments often rely on both private intelligence and their own agencies to attribute cyber incidents to perpetrator groups or sponsor states. If an incident is labelled as the work of Group A by one prominent vendor but Group B by another, international partners might each cite different intelligence sources, muddying the waters of attribution. Adversary nations can exploit these discrepancies for propaganda or denial: a state accused of an attack could point to alternative attributions from respected vendors to cast doubt on the accusation. Thus, inconsistent actor attribution risks undermining the credibility of geopolitical claims. On the other hand, our data shows that when it comes to attributing attacks to a sponsoring country, strategic responses (diplomatic *démarches*, sanctions, or even retaliatory cyber operations) can be decided with greater confidence. In practice, governments seldom rely on vendor reports alone for such decisions, but these reports shape public narratives and ally coordination.

8.2. Recommendations

We outline recommendations for different stakeholders in the threat intelligence ecosystem:

TI Consumers (Defenders). Organizations might be better off adopting a broader, technique-focused approach to defense rather than focusing on actors. TTPs might not be as helpful in attribution as is often assumed [12], but they do provide insight into what techniques might be deployed against their defenses – for example, using MITRE ATT&CK patterns common to multiple actors. If resources allow, use multiple intelligence sources and cross-verify actor attributions rather than relying on one feed. Finally, ensure that compliance teams cross-check any identified actor against official sanction or watch lists using all known aliases, not just the names of their own vendors.

TI Vendors. There are benefits to collaborating on common actor identifiers or at least transparently mapping actor names to industry-recognized aliases [85].

While there are good reasons for initially using vendor-specific aliases [15], [63], the proliferation of over 10 synonyms for the same threat group is a disservice to defenders. Eventually, aliases should be merged.

Vendors should also clarify their focus areas and tracked actors, so clients understand which parts of the actor landscape might not be covered by a given feed.

Improving coverage likely depends on information sharing in the industry, similar to how the anti-virus vendors pooled their malware samples. To some extent, this is undoubtedly already happening, as reflected in vendor reports that explicitly link to the attributions of other vendors. It suggests intelligence sharing via trusted channels [27]. Yet the ability to more comprehensively cover the threat landscape still seems out of reach. To stick with the parallel of the AV industry: any AV solution worth its salt needs to defend against a much wider range of malware than the samples that the vendor is able to collect. So all products improve from pooling samples. There are free-rider risks in such an arrangement [90], [91], but overall it has become the norm. The benefits of sharing have turned out to be stronger than the incentive to monetize exclusive, proprietary data. The TI industry should consider going the same route.

Policymakers and Regulators. Government and international bodies can work with the security industry to maintain an up-to-date public taxonomy of threat groups (building on efforts like MISP TAG or MITRE's group listings) that includes known aliases and supporting evidence of equivalence. This would give organizations a reference point to reconcile vendor information, especially for legal and diplomatic purposes. Finally, governments include intelligence agencies. These have unique methods to support actor attribution. Governments can support collaboration in this nexus. While intelligence agencies cannot cover the width of the private sector, sharing with the private sector would increase the protection of civil entities relying on those commercial products.

8.3. Limitations

Our study has several limitations. First, we do not have ground truth data on actors. This prevents us – and in most cases, everyone else – from knowing whether an attribution is correct. To cope with this limitation, we have focused on (dis)agreement among vendors as a proxy for accuracy.

Second, as we rely on our partner organization, we work with the set of TI vendors that they have contracted with. This makes our findings biased towards a single enterprise partner. Moreover, while these are top-tier TI vendors, as indicated in the fees for their products, no set of seven vendors could claim to represent the bulk of the market. Many firms produce TI, either as their main focus or as a by-product of other security services. Specific numbers across buyers and TI feeds may vary. However, given the absence of scientific reporting on attribution and the unavailability

of expensive commercial TI feeds to researchers, we believe that, despite potential biases, our study still yields valuable insights.

Third, the country agreement we reported may be inflated, as we derived country labels via the TAG mapping, without checking all industry reports. However, during the process of extending TAG, we checked many original TI reports and found no disagreement on the country.

Fourth, we decided to exclude IOCs that led to ambiguous mappings. The actual disagreement of vendors is therefore likely to be higher. We acknowledge this, but we believe that attribution, by its very nature, is mired in uncertainty. We wanted to be conservative and err on the side of agreement by focusing the comparison on the most favorable conditions for reaching agreement.

Fifth, vendor V2 provides many IOCs, many more compared to the other vendors. This introduces a bias where we measure the agreements of other vendors with vendor V2. Nonetheless, other vendors also contribute greatly to the set of co-observed IOCs, so we consider the current approach valuable.

Finally, our augmentation process relies on vendor reports and is a manual task. This may propagate vendor-specific biases and annotator biases. As described in Section 3, vendor reports play a part in the attribution process. We treat this as a validation of our approach and believe the bias to be limited.

8.4. Future Research

This study scratches the surface of actor attribution in TI. We provide three directions for future research. First, we investigated agreement among *commercial* TI vendors. Future work could focus on comparing attribution among public sources and commercial sources versus public sources. One challenge in this research direction is the mapping (process) for actors. Public sources may not have the authoritative reporting that commercial TI vendors provide. Second, we made an initial attempt at explaining root causes for disagreement by looking at the software types for hashes. New work might examine what other root causes underlie disagreement. Mixed-methods studies that collect and analyze quantitative data and interviews may provide valuable insights. Lastly, future work could combine the IOCs from multiple organizations to obtain a broader understanding.

9. Conclusions

We have provided an empirical view on the attribution of 13.5M IOCs by seven leading TI vendors. The gaps we uncovered have impact on the use cases for attributed TI. We explored those implications and reflected on recommendations to improve the status quo.

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Ethics Considerations

We discuss the ethics of our study following the Menlo Report [92].

Respect for Persons. We did not interact with participants directly. We collaborated with a national government organization to access commercial threat intelligence and anonymized both the organization and vendors to prevent reputational or economic harm. Data were aggregated and fields enabling de-anonymization removed. We established explicit agreements on data use and publication. Although vendor-published IOCs included IPv4 addresses, we analyzed these only in aggregate without enrichment, and no PII was exported. The study was conducted with full academic independence.

Beneficence. Research on commercial threat intelligence remains limited. This work benefits CTI stakeholders by providing evidence-based insights into attribution practices and industry limitations. Risks are minimal: vendors are anonymized, operational insights are already publicly accessible, and we coordinated closely with the partner organization.

Justice: Fairness and Equity. The study reflects a collection bias from a single partner organization, but no additional selection or exclusion criteria were applied beyond granted permissions.

Respect for Law and Public Interest. We complied with applicable laws, obtained explicit permission to analyze the data, and conducted VirusTotal lookups internally to avoid data leakage.

Appendix A.

TABLE 5. ACTOR COUNTS IN PUBLIC MAPPINGS

Source	# Actors
MISP TAG	855
BreachHQ	760
Thai CERT	504
MITRE	171
Qianxin	66

TABLE 6. FILETYPES WITH ONLY AGREEMENT

File type	Agree
ZIP	63
JavaScript	35
PDF	22
MS Word Document	14
Email	9
unknown	6
Rich Text Format	5
RAR	3
Mach-O	2
Text	2
DOS batch file	1
PGP Security Key	1
Shell script	1
Office Open XML Presentation	1
Windows Installer	1
INI	1
Compiled HTML Help	1

TABLE 7. FILETYPES WITH ONLY DISAGREEMENT

File type	Disagree
Office Open XML Spreadsheet	1

TABLE 8. AMBIGUOUS ACTOR NAMES IN TAG ($n = 23$)

Name	Name	Name
circle typhoon	evasive panda	andariel
bronze highland	mint sandstorm	sapphire sleet
nobelium	pla navy	g0013
allanite	apt43	greenbug
stardust chollima	iridium	spandex tempest
golden chickens	imperial kitten	smoke sandstorm
cobalt gypsy	cadet blizzard	oilrig
grizzly steppe	cuboid sandstorm	

TABLE 9. CS/MS MAPPING COMPARISON AGAINST TAG

A. Actors not present in TAG	
Amethyst Rain	VolcanicTimber
Highground	DRAGNET PANDA
Houndstooth Typhoon	CHATTY SPIDER
Storm-0252	MintedSoil (CyberRoot)
Wisteria Tsunami	RENEGADE JACKAL
Pinstripe Lightning	
B. Names deconflicted in CS/MS but mapped to different actors in TAG	
LABYRINTH CHOLLIMA	Jade Sleet
Midnight Blizzard	INDRIK SPIDER
Mustard Tempest	VELVET CHOLLIMA
Opal Sleet	LOTUS PANDA
Raspberry Typhoon	VELVET CHOLLIMA
Ruby Sleet	VICE SPIDER
Vanilla Tempest	ALPHA SPIDER
Velvet Tempest	Cozy Bear

TABLE 10. ACTOR AND COUNTRY ATTRIBUTION AGREEMENT FOR 4 VENDORS. VALUES SHOW KRIPPENDORFF'S α WITH 95% CI AND SAMPLE SIZE (n).

Type	OP	4 vendors	
		Actor	Country
ALL	30	1.00 [1.00,1.00] ($n=1$)	1.00 [1.00,1.00] ($n=1$)
ALL	60	1.00 [1.00,1.00] ($n=6$)	1.00 [1.00,1.00] ($n=6$)
ALL (inc. WIP)	14	-0.17 [-0.17,0.00] ($n=2$)	-0.17 [-0.17,0.00] ($n=2$)
ALL (inc. WIP)	30	-0.06 [-0.24,0.07] ($n=9$)	-0.12 [-0.29,0.01] ($n=9$)
ALL (inc. WIP)	60	0.14 [-0.04,0.26] ($n=20$)	0.07 [-0.11,0.21] ($n=20$)
MD5	30	1.00 [1.00,1.00] ($n=1$)	1.00 [1.00,1.00] ($n=1$)
MD5	60	1.00 [1.00,1.00] ($n=5$)	1.00 [1.00,1.00] ($n=5$)
DOMAIN	60	1.00 [1.00,1.00] ($n=1$)	1.00 [1.00,1.00] ($n=1$)

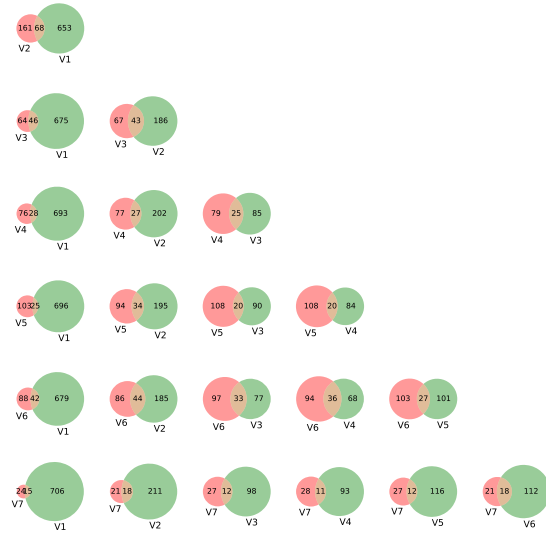


Figure 8. Overlap of tracked actors between vendors.

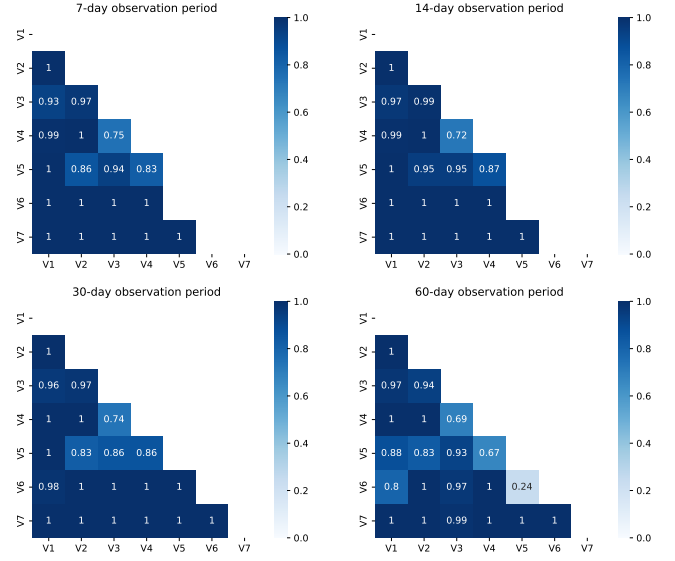


Figure 9. Pairwise vendor agreement on country attribution for co-observed IOCs.

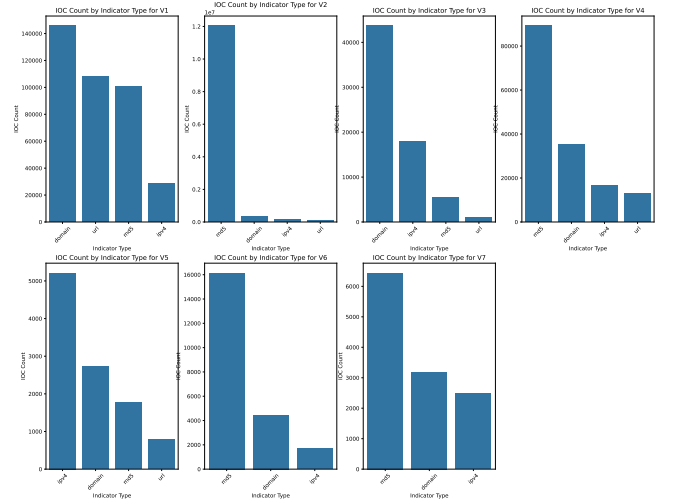


Figure 10. IOC count per indicator type per vendor

TABLE 11. ACTOR AND COUNTRY ATTRIBUTION AGREEMENT FOR ALL INDICATORS (EX. WIP) FOR SECONDARY INDICATOR TYPES. VALUES SHOW KRIPPENDORFF'S α WITH 95% CI AND SAMPLE SIZE (n).

Type	OP (Days)	2 vendors	
		Actor	Country
ALL	7	0.40 [0.33,0.46] ($n=93$)	1.00 [1.00,1.00] ($n=93$)
ALL	14	0.37 [0.32,0.42] ($n=152$)	1.00 [1.00,1.00] ($n=152$)
ALL	30	0.34 [0.30,0.38] ($n=238$)	1.00 [1.00,1.00] ($n=236$)
ALL	60	0.29 [0.26,0.32] ($n=375$)	1.00 [1.00,1.00] ($n=372$)

Appendix B. Meta-Review

The following meta-review was prepared by the program committee for the 2026 IEEE Symposium on Security and Privacy (S&P) as part of the review process as detailed in the call for papers.

B.1. Summary

This paper explores attribution consistency within threat intelligence feeds by comparing indicators of compromise, and associated attributions, across millions of datapoints collected from seven leading vectors over several years.

B.2. Scientific Contributions

- Independent Confirmation of Important Results with Limited Prior Research.
- Provides a Valuable Step Forward in an Established Field.

B.3. Reasons for Acceptance

- 1) The paper provides both independent confirmation of important results with limited prior research, and a valuable step forward in an established field. Prior work has established problems with correctness and consistency in threat intelligence.
- 2) This paper specifically examines problems with attribution, while analyzing a large longitudinal dataset from market-leading threat-intelligence vendors. The quantitative analysis is rigorous.